

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session 98d5acc3 · Data hash d1c5...7edf

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 22:28 and 22:55 UTC. Reported results: test_rmse_mean 0.441733, paper_test_rmse_mean 0.441392. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	98d5acc3-0dad-4424-97ee-52143d7d4ad1
Initialized	2026-10-07T22:28:23Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	bba4a989f0e090b06b8bffc1b347ef051e4b991fa318e0cee5fec2ed42898299
Disclosure	open
Source code, run 1	e858fd0b4e829c642aa49da8022b467f373dc3e1b9bb6871d0c7c9770df98cc7

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	27 minutes 15 secon...	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
test_rmse	0.443899	14 (15 readings)	explicit
paper_test_rmse	0.44022	14 (15 readings)	explicit
test_rmse_std	0.00136971		explicit

Inline claim	Value	Phase	Output line
test_rmse_mean	0.441733		92
paper_test_rmse_mean	0.441392		93

4 Execution evidence

Coherence

Coherent (0.49): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	7200	135.0 s	1.86 GB	94%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 127 tracked files, 17 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	22:28:23	f6896b33e98d2dc3	53
2	phase seed_1	22:29:59	5559a7f724fa40d4	5
3	phase seed_2	22:31:50	162bc07f6533d7e5	5
4	phase seed_3	22:33:35	e96eebfa1c699351	5
5	phase seed_4	22:35:07	5bf1481be6bcc7ad	5
6	phase seed_5	22:37:02	886980c7fcb29678	5
7	phase seed_6	22:39:09	c6eef0fb6acc604f	5
8	phase seed_7	22:41:02	15988ebce8c7718c	5
9	phase seed_8	22:42:55	3fdfc0219191c1dc	5
10	phase seed_9	22:45:09	12c22dedcd38661b	5
11	phase seed_10	22:47:00	cfaf07736692d2fa	5
12	phase seed_11	22:48:36	3b807a106d535ac9	5
13	phase seed_12	22:50:18	43a9dfbc37d3fefdf	5
14	phase seed_13	22:52:06	83d73e100f0587c8	5
15	phase seed_14	22:53:50	706b0b1204d6f0b3	5
16	run_end	22:55:39	cf49fbb54e20742e	4

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 81 files read, 203 files written, 31 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T22:55:40.796955+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	d1c525f2d120c4578b84e3f423dcc756e3d53f5a359fbccef3f573cb066a7edf
Code bundle hash	84a715982fed380fe7bb24beef460e16d65c94a270263a1019a3c8deba654621

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

- [1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
test_rmse	0.442202	0	61	explicit
paper_test_rmse	0.441786	0	62	explicit
test_rmse	0.440959	1	63	explicit
paper_test_rmse	0.442407	1	64	explicit
test_rmse	0.441437	2	65	explicit
paper_test_rmse	0.438213	2	66	explicit
test_rmse	0.441949	3	67	explicit
paper_test_rmse	0.441478	3	68	explicit
test_rmse	0.443906	4	69	explicit
paper_test_rmse	0.442347	4	70	explicit
test_rmse	0.441429	5	71	explicit
paper_test_rmse	0.440706	5	72	explicit
test_rmse	0.443826	6	73	explicit
paper_test_rmse	0.442787	6	74	explicit
test_rmse	0.441378	7	75	explicit
paper_test_rmse	0.440095	7	76	explicit
test_rmse	0.438714	8	77	explicit
paper_test_rmse	0.442087	8	78	explicit
test_rmse	0.441689	9	79	explicit
paper_test_rmse	0.441253	9	80	explicit
test_rmse	0.441171	10	81	explicit
paper_test_rmse	0.441882	10	82	explicit
test_rmse	0.441219	11	83	explicit
paper_test_rmse	0.441034	11	84	explicit
test_rmse	0.441748	12	85	explicit
paper_test_rmse	0.441756	12	86	explicit
test_rmse	0.440467	13	87	explicit
paper_test_rmse	0.442827	13	88	explicit
test_rmse	0.443899	14	89	explicit
paper_test_rmse	0.44022	14	90	explicit
test_rmse_std	0.00136971		91	explicit

Phase	Line	Time	CPU time	Memory	GPU
seed_0	1	22:28:24	0.4 s	0.44 GB	
seed_1	5	22:29:59	0.1 s	0.02 GB	
seed_2	9	22:31:50	0.0 s	0.03 GB	
seed_3	13	22:33:35	0.1 s	0.03 GB	
seed_4	17	22:35:07	0.0 s	0.02 GB	
seed_5	21	22:37:02	0.0 s	0.03 GB	
seed_6	25	22:39:09	0.0 s	0.02 GB	
seed_7	29	22:41:02	0.1 s	0.03 GB	
seed_8	33	22:42:55	0.1 s	0.02 GB	
seed_9	37	22:45:09	0.0 s	0.02 GB	
seed_10	41	22:47:00	0.0 s	0.02 GB	

Phase	Line	Time	CPU time	Memory	GPU
seed_11	45	22:48:36	0.0 s	0.02 GB	
seed_12	49	22:50:18	0.1 s	0.03 GB	
seed_13	53	22:52:06	0.1 s	0.02 GB	
seed_14	57	22:53:50	0.1 s	0.02 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	22:28:23	f6896b33e98d2dc3	53
1	phase seed_0	22:28:24	f6896b33e98d2dc3	0
2	phase seed_1	22:29:59	5559a7f724fa40d4	5
3	phase seed_2	22:31:50	162bc07f6533d7e5	5
4	phase seed_3	22:33:35	e96eebfa1c699351	5
5	phase seed_4	22:35:07	5bf1481be6bcc7ad	5
6	phase seed_5	22:37:02	886980c7fcb29678	5
7	phase seed_6	22:39:09	c6eef0fb6acc604f	5
8	phase seed_7	22:41:02	15988ebce8c7718c	5
9	phase seed_8	22:42:55	3fdfc0219191c1dc	5
10	phase seed_9	22:45:09	12c22dedcd38661b	5
11	phase seed_10	22:47:00	cfaf07736692d2fa	5
12	phase seed_11	22:48:36	3b807a106d535ac9	5
13	phase seed_12	22:50:18	43a9dfbc37d3fefc	5
14	phase seed_13	22:52:06	83d73e100f0587c8	5
15	phase seed_14	22:53:50	706b0b1204d6f0b3	5
16	run_end	22:55:39	cf49fbb54e20742e	4

Event	Path or command	Hash
read	kv_run.sh	
read	paper/exp/tabm/california/0-evaluation/0.toml	
read	paper/bin/model.py	
read	.kveritasignore	
read	example.ipynb	
read	paper/bin/ensemble.py	
read	paper/bin/evaluate.py	
read	paper/bin/go.py	
read	paper/bin/tune.py	
read	paper/data/california/info.json	
read	paper/environment.yaml	
read	paper/exp/kv-reproduce/0.toml	
read	paper/exp/tabm/california/0-evaluation/0/report.json	
read	paper/exp/tabm/california/0-evaluation/1.toml	
read	paper/exp/tabm/california/0-evaluation/1/report.json	
read	paper/exp/tabm/california/0-evaluation/10.toml	
read	paper/exp/tabm/california/0-evaluation/10/report.json	
read	paper/exp/tabm/california/0-evaluation/11.toml	
read	paper/exp/tabm/california/0-evaluation/11/report.json	
read	paper/exp/tabm/california/0-evaluation/12.toml	
read	paper/exp/tabm/california/0-evaluation/12/report.json	

Event	Path or command	Hash
read	paper/exp/tabm/california/0-evaluation/13.toml	
read	paper/exp/tabm/california/0-evaluation/13/report.json	
read	paper/exp/tabm/california/0-evaluation/14.toml	
read	paper/exp/tabm/california/0-evaluation/14/report.json	
read	paper/exp/tabm/california/0-evaluation/2.toml	
read	paper/exp/tabm/california/0-evaluation/2/report.json	
read	paper/exp/tabm/california/0-evaluation/3.toml	
read	paper/exp/tabm/california/0-evaluation/3/report.json	
read	paper/exp/tabm/california/0-evaluation/4.toml	
read	paper/exp/tabm/california/0-evaluation/4/report.json	
read	paper/exp/tabm/california/0-evaluation/5.toml	
read	paper/exp/tabm/california/0-evaluation/5/report.json	
read	paper/exp/tabm/california/0-evaluation/6.toml	
read	paper/exp/tabm/california/0-evaluation/6/report.json	
read	paper/exp/tabm/california/0-evaluation/7.toml	
read	paper/exp/tabm/california/0-evaluation/7/report.json	
read	paper/exp/tabm/california/0-evaluation/8.toml	
read	paper/exp/tabm/california/0-evaluation/8/report.json	
read	paper/exp/tabm/california/0-evaluation/9.toml	
read	paper/exp/tabm/california/0-evaluation/9/report.json	
read	paper/lib/__init__.py	
read	paper/lib/deep.py	
read	paper/pixi.toml	
read	paper/pyproject.toml	
read	paper/tools/prepare_tabred.py	
read	pyproject.toml	
read	tabm.py	
read	tabm_reference.py	
read	test_code_blocks.py	
read	paper/cache/build_dataset__california__ab1828c895a3404e62f53860d501bb2...	
write	paper/exp/kv-reproduce/0/events.out.tfevents.1791412106.localhost.311504.0	5157ceee41d96d73
write	paper/exp/kv-reproduce/0/loss_train/events.out.tfevents.1791412107.localhos...	1a4767f0030dd389
write	paper/exp/kv-reproduce/0/score_test/events.out.tfevents.1791412107.localhos...	8b7cd5b4329738c2
write	paper/exp/kv-reproduce/0/score_val/events.out.tfevents.1791412107.localhost...	01e185958649ce06
write	paper/exp/kv-reproduce/0/predictions.npz	4085303ce33131ae
write	paper/exp/kv-reproduce/0/summary.json	de491bf3e693f2ef
write	paper/exp/kv-reproduce/0/checkpoint.pt	e470e19b5819aaff
write	paper/exp/kv-reproduce/0-greedy-heads/report.json	b7c64c9e67cb6b72
write	paper/exp/kv-reproduce/0-greedy-heads/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/0/report.json	6fd74d5221b0e257
write	paper/exp/kv-reproduce/0/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/1.toml	e91d624d5b620b66
read	paper/exp/kv-reproduce/0-best-head/report.json	
read	paper/lib/env.py	
read	paper/lib/util.py	
read	paper/lib/data.py	
read	paper/lib/metrics.py	
write	paper/exp/kv-reproduce/1/events.out.tfevents.1791412202.localhost.314853.0	b430afda24342e11
write	paper/exp/kv-reproduce/1/norm_output.weight/events.out.tfevents.179141220...	5ffaaed7b5f5902a
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.0.0.bias/events.out.tfeve...	4fce278940379819

Event	Path or command	Hash
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.1.0.s/events.out.tfevents.17...	413524c1ad9482f6
write	paper/exp/kv-reproduce/1/norm_output.bias/events.out.tfevents.1791412202.l...	6a58b1ada5517e45
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.1.0.s/events.out.tfevent...	8b4f33b122bded2f
write	paper/exp/kv-reproduce/1/gradnorm_output.weight/events.out.tfevents.17914...	8ad016002a3ad309
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.1.0.r/events.out.tfevents.17...	f03408e5fdbf098b
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.0.0.r/events.out.tfevent...	2d1f6be9d97f57d2
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.1.0.weight/events.out.tf...	8067ea75f622497f
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.0.0.bias/events.out.tfevents....	d479cdc27974a4da
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.0.0.r/events.out.tfevents.17...	649499643b3efacf
write	paper/exp/kv-reproduce/1/gradnorm_model/events.out.tfevents.1791412202.l...	d03314f59c155b6d
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.2.0.bias/events.out.tfeve...	f52877772e09d697
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.2.0.r/events.out.tfevents.17...	3414060197386c94
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.2.0.s/events.out.tfevent...	10d34094c1381f25
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.2.0.s/events.out.tfevents.17...	a18cc010c7212ed7
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.0.0.s/events.out.tfevent...	cf09057511013391
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.1.0.r/events.out.tfevent...	7c7e2a78fd3a1c5c
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.1.0.bias/events.out.tfev...	27430dee14df6845
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.2.0.r/events.out.tfevent...	93f743543ebd28a5
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.2.0.bias/events.out.tfev...	9c3a0161635c9af4
write	paper/exp/kv-reproduce/1/gradratio_output.weight/events.out.tfevents.179141...	c4286404ada70bdf
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.2.0.s/events.out.tfevents...	37a1c3b4d594acd1
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.1.0.s/events.out.tfevents...	221a2a32b7ec2b0c
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.1.0.weight/events.out.tf...	86c5e3f809c9aea7
write	paper/exp/kv-reproduce/1/gradratio_output.bias/events.out.tfevents.17914122...	ce3187fef6d1f0a8
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.0.0.r/events.out.tfevents...	76613ef5786f52e9
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.0.0.s/events.out.tfevents.17...	959ef1a8ced84b2b
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.2.0.weight/events.out.tfeve...	c408a528b9bd45ec
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.0.0.bias/events.out.tfev...	3841a77d1332eb41
write	paper/exp/kv-reproduce/1/gradnorm_output.bias/events.out.tfevents.1791412...	4d4e1ac10af8b3f4
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.0.0.weight/events.out.tfeve...	1733884b8d2847e3
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.0.0.weight/events.out.tf...	f2d405b45d98c489
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.2.0.r/events.out.tfevents...	804552450dfd9e2d
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.1.0.weight/events.out.tfeve...	2a2dc5ae7d06e6f3
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.0.0.s/events.out.tfevents...	56a7e68f6d2629f4
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.0.0.weight/events.out.tf...	ee18eef27d76975e
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.1.0.bias/events.out.tfeve...	1da5d7c4b90591c1
write	paper/exp/kv-reproduce/1/gradnorm_backbone.blocks.2.0.weight/events.out.tf...	ec3dfd3e60523ad6
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.1.0.bias/events.out.tfevents....	47e9dabc6f2f4491
write	paper/exp/kv-reproduce/1/norm_model/events.out.tfevents.1791412202.local...	8389cc6c054f534e
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.1.0.r/events.out.tfevents...	e9372eb8a2d49d4f
write	paper/exp/kv-reproduce/1/norm_backbone.blocks.2.0.bias/events.out.tfevents....	a280c13102bcf9b7
write	paper/exp/kv-reproduce/1/gradratio_backbone.blocks.2.0.weight/events.out.tf...	5af140f69c30d6a9
write	paper/exp/kv-reproduce/1/score_val/events.out.tfevents.1791412204.localhost...	02a7604ac4bac84c
write	paper/exp/kv-reproduce/1/score_test/events.out.tfevents.1791412204.localhos...	b3ee42299e8c7d0c
write	paper/exp/kv-reproduce/1/loss_train/events.out.tfevents.1791412204.localhos...	6310346c67680aa4
write	paper/exp/kv-reproduce/1/predictions.npz	023d90b130fedf26
write	paper/exp/kv-reproduce/1/summary.json	de2e323298efb975
write	paper/exp/kv-reproduce/1/checkpoint.pt	8e6bf3ad3cddc329
write	paper/exp/kv-reproduce/1/report.json	ade9a01763562612
write	paper/exp/kv-reproduce/1/DONE	e3b0c44298fc1c14

Event	Path or command	Hash
write	paper/exp/kv-reproduce/2.toml	6406d8e4d29d3307
read	paper/exp/kv-reproduce/1-best-head/report.json	
read	paper/exp/kv-reproduce/1-greedy-heads/report.json	
write	paper/exp/kv-reproduce/2/events.out.tfevents.1791412312.localhost.318729.0	6736152fd9930dcf
write	paper/exp/kv-reproduce/2/score_val/events.out.tfevents.1791412313.localhost...	67b369ad30206f3f
write	paper/exp/kv-reproduce/2/loss_train/events.out.tfevents.1791412313.localhos...	056190f0a3e61606
write	paper/exp/kv-reproduce/2/score_test/events.out.tfevents.1791412313.localhos...	daf9031899b43087
write	paper/exp/kv-reproduce/2/predictions.npz	e8bbca389833a085
write	paper/exp/kv-reproduce/2/summary.json	65efe4196767f9ae
write	paper/exp/kv-reproduce/2/checkpoint.pt	62352cfd23cd43e2
write	paper/exp/kv-reproduce/2/report.json	16ce491a1dddd600
write	paper/exp/kv-reproduce/2/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/3.toml	f07f3ea589a2ae6f
read	paper/exp/kv-reproduce/2-best-head/report.json	
read	paper/exp/kv-reproduce/2-greedy-heads/report.json	
write	paper/exp/kv-reproduce/3/events.out.tfevents.1791412417.localhost.322407.0	af119c61ae408d30
write	paper/exp/kv-reproduce/3/score_val/events.out.tfevents.1791412418.localhost...	d6b864c0ede11f52
write	paper/exp/kv-reproduce/3/loss_train/events.out.tfevents.1791412418.localhos...	212fc422c57bb517
write	paper/exp/kv-reproduce/3/score_test/events.out.tfevents.1791412418.localhos...	c878a40ffd21e357
write	paper/exp/kv-reproduce/3/predictions.npz	364c6b12ae65d54f
write	paper/exp/kv-reproduce/3/summary.json	014605203f79b9a8
write	paper/exp/kv-reproduce/3/checkpoint.pt	0e3db3427532c67e
write	paper/exp/kv-reproduce/3/report.json	578274f1551b80fa
write	paper/exp/kv-reproduce/3/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/4.toml	73880757a7418222
read	paper/exp/kv-reproduce/3-best-head/report.json	
read	paper/exp/kv-reproduce/3-greedy-heads/report.json	
write	paper/exp/kv-reproduce/4/events.out.tfevents.1791412509.localhost.325608.0	29317a3d3cfbd2a8
write	paper/exp/kv-reproduce/4/loss_train/events.out.tfevents.1791412510.localhos...	488b8d4702618095
write	paper/exp/kv-reproduce/4/score_val/events.out.tfevents.1791412510.localhost...	44868b14169d9b22
write	paper/exp/kv-reproduce/4/score_test/events.out.tfevents.1791412510.localhos...	c3ecd023585ebd6b
write	paper/exp/kv-reproduce/4/predictions.npz	f82cacf197cac631
write	paper/exp/kv-reproduce/4/summary.json	cd40d74277f75b39
write	paper/exp/kv-reproduce/4/checkpoint.pt	f061d5a2e3f40327
write	paper/exp/kv-reproduce/4/report.json	e5afa8aab56eaa6c
write	paper/exp/kv-reproduce/4/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/5.toml	09f8b4fe77605679
read	paper/exp/kv-reproduce/4-best-head/report.json	
read	paper/exp/kv-reproduce/4-greedy-heads/report.json	
write	paper/exp/kv-reproduce/5/events.out.tfevents.1791412625.localhost.329654.0	10a062f9b5f359c3
write	paper/exp/kv-reproduce/5/loss_train/events.out.tfevents.1791412626.localhos...	8927096bb993b1b7
write	paper/exp/kv-reproduce/5/score_val/events.out.tfevents.1791412626.localhost...	189ce6844ced57a7
write	paper/exp/kv-reproduce/5/score_test/events.out.tfevents.1791412626.localhos...	a0d608e8016dadcd
write	paper/exp/kv-reproduce/5/predictions.npz	b314419849179116
write	paper/exp/kv-reproduce/5/summary.json	cc9d9cd8e88c10dc
write	paper/exp/kv-reproduce/5/checkpoint.pt	a734742d48c00382
write	paper/exp/kv-reproduce/5-best-head/report.json	c0b4bf0f669d0e0a
write	paper/exp/kv-reproduce/5-best-head/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/5/report.json	128a87b3ec5ab13f
write	paper/exp/kv-reproduce/5/DONE	e3b0c44298fc1c14

Event	Path or command	Hash
write	paper/exp/kv-reproduce/6.toml	83bae43609d19bba
read	paper/exp/kv-reproduce/5-greedy-heads/report.json	
write	paper/exp/kv-reproduce/6/events.out.tfevents.1791412751.localhost.334054.0	a89a532826c70662
write	paper/exp/kv-reproduce/6/loss_train/events.out.tfevents.1791412753.localhos...	06e84ea312c3bfcf
write	paper/exp/kv-reproduce/6/score_test/events.out.tfevents.1791412753.localhos...	139c7a1993b09ee7
write	paper/exp/kv-reproduce/6/score_val/events.out.tfevents.1791412753.localhost...	9f3a567327700ba2
write	paper/exp/kv-reproduce/6/predictions.npz	c4424f047cb9fd4f
write	paper/exp/kv-reproduce/6/summary.json	0255c14aa22921b8
write	paper/exp/kv-reproduce/6/checkpoint.pt	3bc3770071fed271
write	paper/exp/kv-reproduce/6/report.json	525fe7f79d641048
write	paper/exp/kv-reproduce/6/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/7.toml	91c4f8adb79171a3
read	paper/exp/kv-reproduce/6-best-head/report.json	
read	paper/exp/kv-reproduce/6-greedy-heads/report.json	
write	paper/exp/kv-reproduce/7/events.out.tfevents.1791412864.localhost.337981.0	df7ffebe1cab81ed
write	paper/exp/kv-reproduce/7/score_test/events.out.tfevents.1791412865.localhos...	fdb232bf78f77bd5
write	paper/exp/kv-reproduce/7/score_val/events.out.tfevents.1791412865.localhost...	fab958df94876806
write	paper/exp/kv-reproduce/7/loss_train/events.out.tfevents.1791412865.localhos...	28ba9d703d492a06
write	paper/exp/kv-reproduce/7/predictions.npz	344950a541a4811a
write	paper/exp/kv-reproduce/7/summary.json	07554a4a0bb39cf9
write	paper/exp/kv-reproduce/7/checkpoint.pt	97c0993812476a4a
write	paper/exp/kv-reproduce/7/report.json	d4423bd3d79b436e
write	paper/exp/kv-reproduce/7/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/8.toml	0f65852024da14c1
read	paper/exp/kv-reproduce/7-best-head/report.json	
read	paper/exp/kv-reproduce/7-greedy-heads/report.json	
write	paper/exp/kv-reproduce/8/events.out.tfevents.1791412977.localhost.341929.0	2dc8f53ca6e167fc
write	paper/exp/kv-reproduce/8/score_val/events.out.tfevents.1791412978.localhost...	f84f8956332f374e
write	paper/exp/kv-reproduce/8/loss_train/events.out.tfevents.1791412978.localhos...	69389c98e6f96229
write	paper/exp/kv-reproduce/8/score_test/events.out.tfevents.1791412978.localhos...	d1d2921babe38303
write	paper/exp/kv-reproduce/8/predictions.npz	c43daefa48ac2987
write	paper/exp/kv-reproduce/8/summary.json	3c9df592c80a8802
write	paper/exp/kv-reproduce/8/checkpoint.pt	3ea021168347d11f
write	paper/exp/kv-reproduce/8/report.json	6ea3b358f75d3ce3
write	paper/exp/kv-reproduce/8/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/9.toml	cf29d8fdf1249770
read	paper/exp/kv-reproduce/8-best-head/report.json	
read	paper/exp/kv-reproduce/8-greedy-heads/report.json	
write	paper/exp/kv-reproduce/9/events.out.tfevents.1791413111.localhost.346599.0	7db5af3330c51c11
write	paper/exp/kv-reproduce/9/score_test/events.out.tfevents.1791413112.localhos...	ed81a0867c7cfee9
write	paper/exp/kv-reproduce/9/score_val/events.out.tfevents.1791413112.localhost...	c50bd7926180cd36
write	paper/exp/kv-reproduce/9/loss_train/events.out.tfevents.1791413112.localhos...	f9dc0ed5c1f6d760
write	paper/exp/kv-reproduce/9/predictions.npz	fd57b81dec3ff8dd
write	paper/exp/kv-reproduce/9/summary.json	d4d77357d5709112
write	paper/exp/kv-reproduce/9/checkpoint.pt	acee23c266985308
write	paper/exp/kv-reproduce/9-best-head/report.json	6d530ecdc7e3c367
write	paper/exp/kv-reproduce/9-best-head/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/9-greedy-heads/report.json	249e09389ffde43e
write	paper/exp/kv-reproduce/9-greedy-heads/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/9/report.json	7db9e37124ae1003

Event	Path or command	Hash
write	paper/exp/kv-reproduce/9/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/10.toml	bdb560b35a0a3075
write	paper/exp/kv-reproduce/10/events.out.tfevents.1791413223.localhost.350475.0	40c0ebec83c2a80d
read	.gitignore	
write	paper/exp/kv-reproduce/10/score_test/events.out.tfevents.1791413224.localho...	1fe5261ca3a534f3
write	paper/exp/kv-reproduce/10/loss_train/events.out.tfevents.1791413224.localho...	9ce673203d6ba8f9
write	paper/exp/kv-reproduce/10/score_val/events.out.tfevents.1791413224.localho...	fd42ce9476c4e3b4
write	paper/exp/kv-reproduce/10/predictions.npz	e780c066b5bfb113
write	paper/exp/kv-reproduce/10/summary.json	be887727bf3277d0
write	paper/exp/kv-reproduce/10/checkpoint.pt	9d6c8353f5c4fe36
write	paper/exp/kv-reproduce/10/report.json	010acac93b2adab9
write	paper/exp/kv-reproduce/10/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/11.toml	40c8c61bf0821654
read	paper/exp/kv-reproduce/10-best-head/report.json	
read	paper/exp/kv-reproduce/10-greedy-heads/report.json	
write	paper/exp/kv-reproduce/11/events.out.tfevents.1791413318.localhost.353854.0	112481d523fcc51f
write	paper/exp/kv-reproduce/11/score_val/events.out.tfevents.1791413319.localho...	35923e9b0fbef7f7
write	paper/exp/kv-reproduce/11/score_test/events.out.tfevents.1791413319.localho...	b388b2d502691954
write	paper/exp/kv-reproduce/11/loss_train/events.out.tfevents.1791413319.localho...	c52fcb98e4a6e704
write	paper/exp/kv-reproduce/11/predictions.npz	9f8eb9414a409fba
write	paper/exp/kv-reproduce/11/summary.json	1d0001bd73084575
write	paper/exp/kv-reproduce/11/checkpoint.pt	cc6e7de7f7c87187
write	paper/exp/kv-reproduce/11/report.json	38124066b98e8f21
write	paper/exp/kv-reproduce/11/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/12.toml	208c3ceab8f20ec7
read	paper/exp/kv-reproduce/11-best-head/report.json	
read	paper/exp/kv-reproduce/11-greedy-heads/report.json	
write	paper/exp/kv-reproduce/12/events.out.tfevents.1791413420.localhost.357434.0	4257b6e06b62a712
write	paper/exp/kv-reproduce/12/score_val/events.out.tfevents.1791413421.localho...	f598437c749e44a5
write	paper/exp/kv-reproduce/12/loss_train/events.out.tfevents.1791413421.localho...	a0111f625305a987
write	paper/exp/kv-reproduce/12/score_test/events.out.tfevents.1791413421.localho...	c31be4e139f3e9b5
write	paper/exp/kv-reproduce/12/predictions.npz	8efb56374c95b092
write	paper/exp/kv-reproduce/12/summary.json	f85469bde4d64e8f
write	paper/exp/kv-reproduce/12/checkpoint.pt	1aee97d3388d8353
write	paper/exp/kv-reproduce/12/report.json	6840b57ba2e2d6d5
write	paper/exp/kv-reproduce/12/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/13.toml	2a456f09509374cb
read	paper/exp/kv-reproduce/12-best-head/report.json	
read	paper/exp/kv-reproduce/12-greedy-heads/report.json	
write	paper/exp/kv-reproduce/13/events.out.tfevents.1791413528.localhost.361190.0	296a77a8a70303f6
write	paper/exp/kv-reproduce/13/score_test/events.out.tfevents.1791413529.localho...	ef8371f2b49c5158
write	paper/exp/kv-reproduce/13/score_val/events.out.tfevents.1791413529.localho...	0e732c3d7595e305
write	paper/exp/kv-reproduce/13/loss_train/events.out.tfevents.1791413529.localho...	47092e4192453e02
write	paper/exp/kv-reproduce/13/predictions.npz	3c44296c8c5cc120
write	paper/exp/kv-reproduce/13/summary.json	c12bbd6ef26794d5
write	paper/exp/kv-reproduce/13/checkpoint.pt	a16ef8adbe80e12c
write	paper/exp/kv-reproduce/13-best-head/report.json	437643dfc226d507
write	paper/exp/kv-reproduce/13-best-head/DONE	e3b0c44298fc1c14
write	paper/exp/kv-reproduce/13/report.json	fcfe85f57ae84d10
write	paper/exp/kv-reproduce/13/DONE	e3b0c44298fc1c14

Event	Path or command	Hash
write	paper/exp/kv-reproduce/14.toml	9ffea84af8dd845a
read	paper/exp/kv-reproduce/13-greedy-heads/report.json	
write	paper/exp/kv-reproduce/14/events.out.tfevents.1791413632.localhost.364828.0	8aa0c26021be5447
write	paper/exp/kv-reproduce/14/score_val/events.out.tfevents.1791413633.localho...	1b3f5471d39662b3
write	paper/exp/kv-reproduce/14/score_test/events.out.tfevents.1791413633.localho...	d9d90e7ef58fa37c
write	paper/exp/kv-reproduce/14/loss_train/events.out.tfevents.1791413633.localho...	5479b1dde35c859d
write	paper/exp/kv-reproduce/14/predictions.npz	ffe0449e971905a0
write	paper/exp/kv-reproduce/14/summary.json	8edf2ee454e031d0
write	paper/exp/kv-reproduce/14/checkpoint.pt	d9090a2ac896b385
write	paper/exp/kv-reproduce/14/report.json	151f6dcaf0d53506
write	paper/exp/kv-reproduce/14/DONE	e3b0c44298fc1c14
read	paper/exp/kv-reproduce/14-best-head/report.json	
read	paper/exp/kv-reproduce/14-greedy-heads/report.json	
process	python bin/model.py exp/kv-reproduce/0.toml --force	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/1.toml --force	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/2.toml --force	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/3.toml --force	
process	tail -3	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/4.toml --force	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/5.toml --force	
process	python bin/model.py exp/kv-reproduce/6.toml --force	
process	tail -3	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/7.toml --force	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/8.toml --force	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/9.toml --force	
process	python bin/model.py exp/kv-reproduce/10.toml --force	
process	tail -3	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/11.toml --force	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/12.toml --force	
process	python bin/model.py exp/kv-reproduce/13.toml --force	
process	tail -3	
process	tail -3	
process	python bin/model.py exp/kv-reproduce/14.toml --force	
process	python - exp/tabm/california/0-evaluation exp/kv-reproduce	

C Run history

Invocation 1, OK, 27 minutes 15 seconds, 93 output lines, started 2026-10-07T22:28:23

```
metric_hash a139da5c23adf13232623c151bcb58b7506054528264f8b5dad5be6241a1238
run_digest e36eeda48b32113ffe1b7f14ed244c5c7b5737189ed3d8e46dc26c859bdec00d
anchored at run end
```

D Raw seal fields and hashes

Seal

```
data_hash d1c525f2d120c4578b84e3f423dcc756e3d53f5a359fbecf3f573cb066a7edf
signed_message_hash 2bd6747eae7fa5fe9429cb0f21bb39bf7cd7348dee3dab8e4120e80467eadfa4
nonce 97840d0e55f799f3302b50b8f2a8f066
signed_at 2026-10-07T22:55:40.796955+00:00
sealed_at 2026-10-07T22:55:40.861328213Z
checkout_bundle 84a715982fed380fe7bb24beef460e16d65c94a270263a1019a3c8deba654621
```

Signature (base64)

```
F7P/DJI2Ee7JRRMZy3LSKRc6T8Au3V2BH1piHaLQZ5mus3i5NooT9U0tA0SX0AHATIymL27DnoW08gfVHq8GELFz
tF0I7HNxemII0QZElpvsXhLcdqT4jxT/eBJ1ZRHb60tL0B+yzISPI7hdhg06EGkaFo1xoVB3vXBtZHEgZ6Xlz7dX
KJ56P20jsXSjjJZVG80a4gnu03STH1CtLdI1woYGNhTgumZa40UDsYJxNA2wfAghXEgMpXbeezJYnh9NCchhd4BK
T0KdtC5qK6HrCvUjQWY5prGqBz9n5wDJKYBG9fSSXktDaNRcjFmL5JnRTN1NIVkN2U9Cslyf2uXUQaXZuBcbprbB
RQcoIkd9bDYfgvmuYwnNVBGIf89apYpKM4EtKA+q7nU+Q7FP0i5ULBmkKcjXLFH+uF+sT4Y+0VbVwIJC2XIMwgdT
QGUVVrvCzXz15uVzSd78973xaHRGnrK1f/dd/MnGn0A9ethxnGJ1C8hQlkysVGdYRMvM0DaMXU6bF/6cssUhXfCRK
aqlfMHYYznmbIJIUWq5Ve44+HDtd/3Uh7GLc6a00pZyDQrmWLGxhHaf32H+UWCQijzi05tG7l0HQ90Hb8mwwA0n4J
tUvoEpHGI9viIUI9u+8WfQ2Q0HgziBwwI8e3EHoKR7vPZ1c9+genBRN/0KBYyg3fWKI=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICijANBgkqhkiG9w0BAQEFAAACAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAt4vcgWP+iWscPc3HzpmjpsKbTmOemqgp6RB6GNNPpxYa5X
7qU95EfEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeAdbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
AhfTRuJNgWEkjdW5lb0Vhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqes7VSzt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMGicNgeNHNCy9bLgpU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/oLCHBCf1VIDCKDxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdw3gHLcCAwEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout 10fc6ca359d4a152938666bb87896e50019b200a1237f181d9a48f02f5c6a0cf
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics a139da5c23adf13232623c151bcb58b7506054528264f8b5dad5be6241a1238
digest e36eeda48b32113ffe1b7f14ed244c5c7b5737189ed3d8e46dc26c859bdec00d
kv_run.sh (unchanged)
pre d9abbdafd98bcb266a6a8ddc3fb34a00b09df53c1951a186a53a634cf991887
post d9abbdafd98bcb266a6a8ddc3fb34a00b09df53c1951a186a53a634cf991887
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,  
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},  
  author={Mamadou K. Keita and Christopher Homan},  
  year={2026},  
  eprint={2605.08586},  
  archivePrefix={arXiv},  
  primaryClass={cs.CR},  
  url={https://arxiv.org/abs/2605.08586},  
}
```